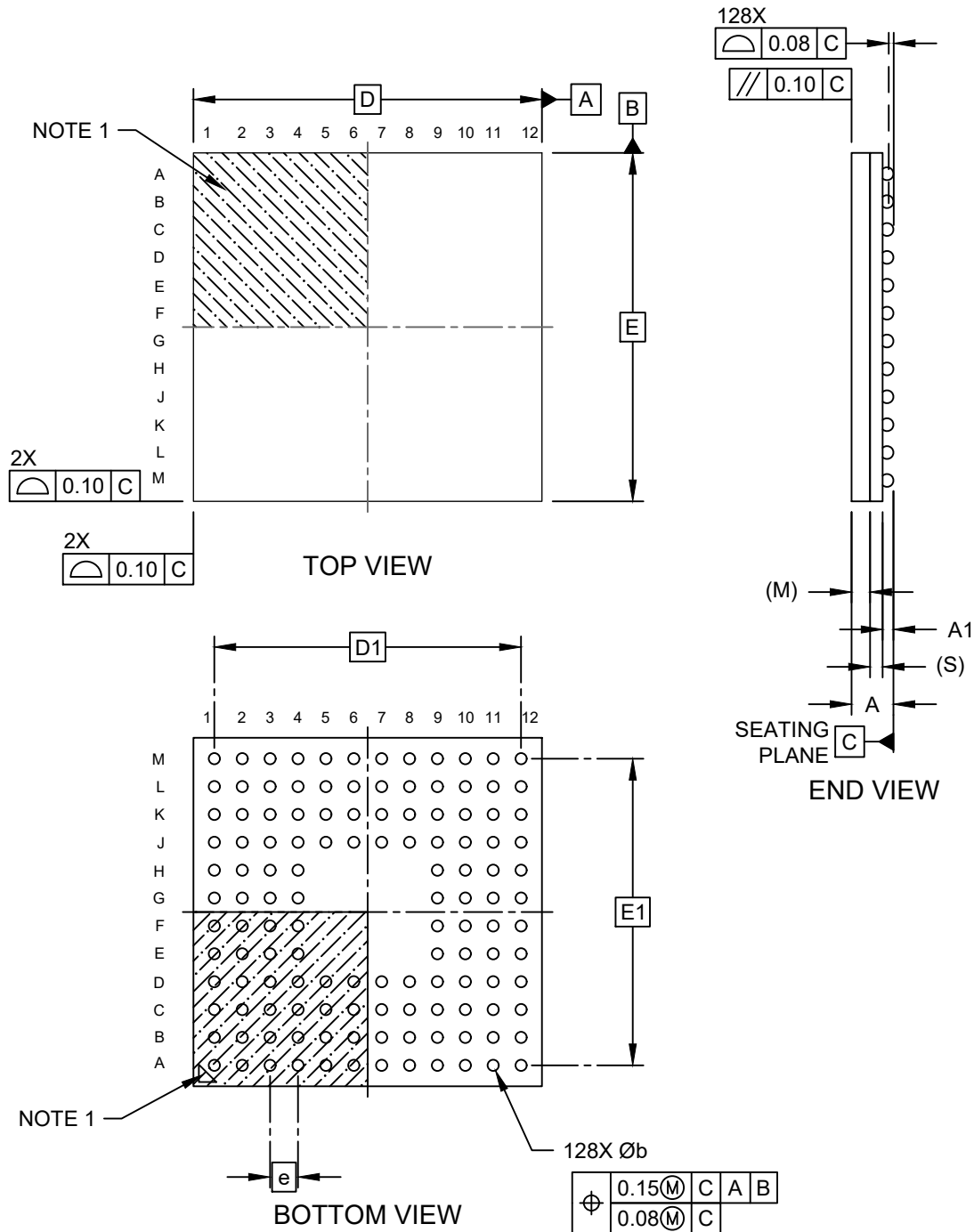


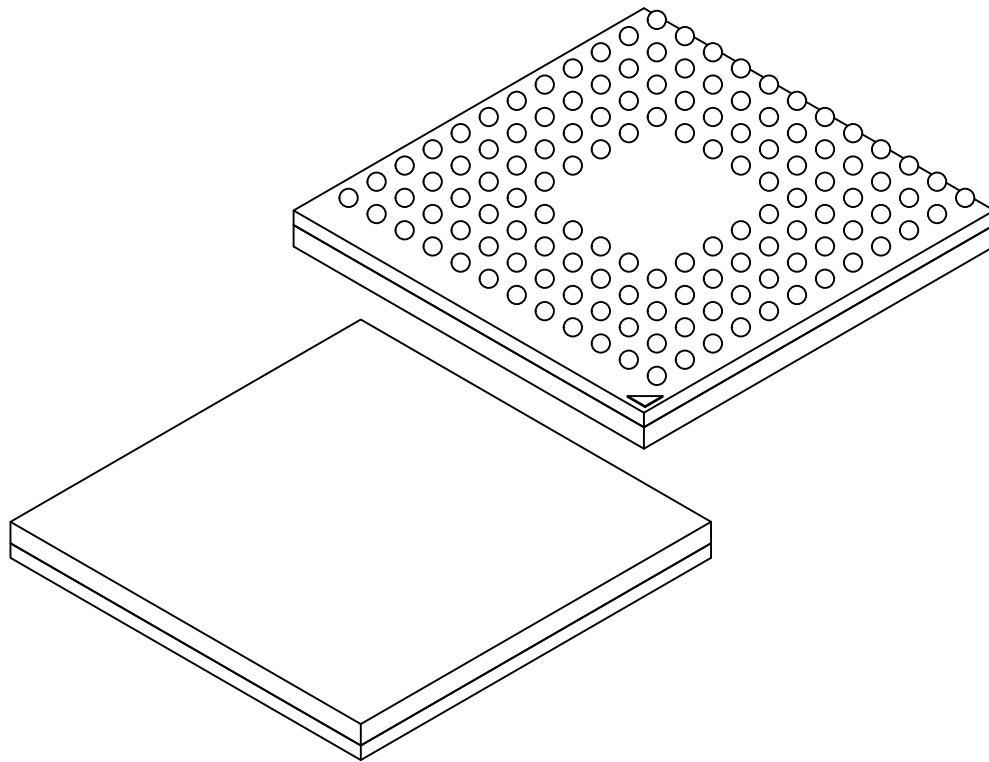
128-Ball Thin Fine-Pitch Ball Grid Array Package (B5B) - 10x10x1.2mm Body [TFBGA] Atmel Legacy GPC: CJZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



128-Ball Thin Fine-Pitch Ball Grid Array Package (B5B) - 10x10x1.2mm Body [TFBGA] Atmel Legacy GPC: CJZ

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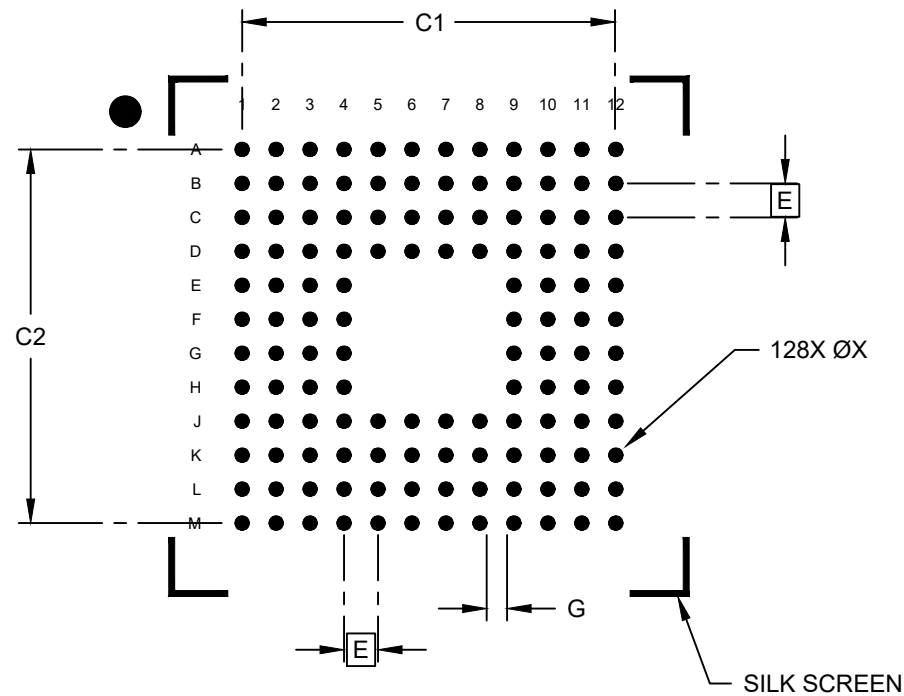
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	128		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.16	–	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.36 REF		
Overall Length	D	10.00 BSC		
Ball Array Length	D2	8.80 BSC		
Overall Width	E	10.00 BSC		
Ball Array Width	E2	8.80 BSC		
Ball Width	b	0.270	–	0.370

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

128-Ball Thin Fine-Pitch Ball Grid Array Package (B5B) - 10x10x1.2mm Body [TFBGA] Atmel Legacy GPC: CJZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Optional Pad Diameter (X128)	X			0.32
Contact Pad Spacing	C1		8.80	
Contact Pad Spacing	C2		8.80	
Column and Row Spacing	G	0.48		

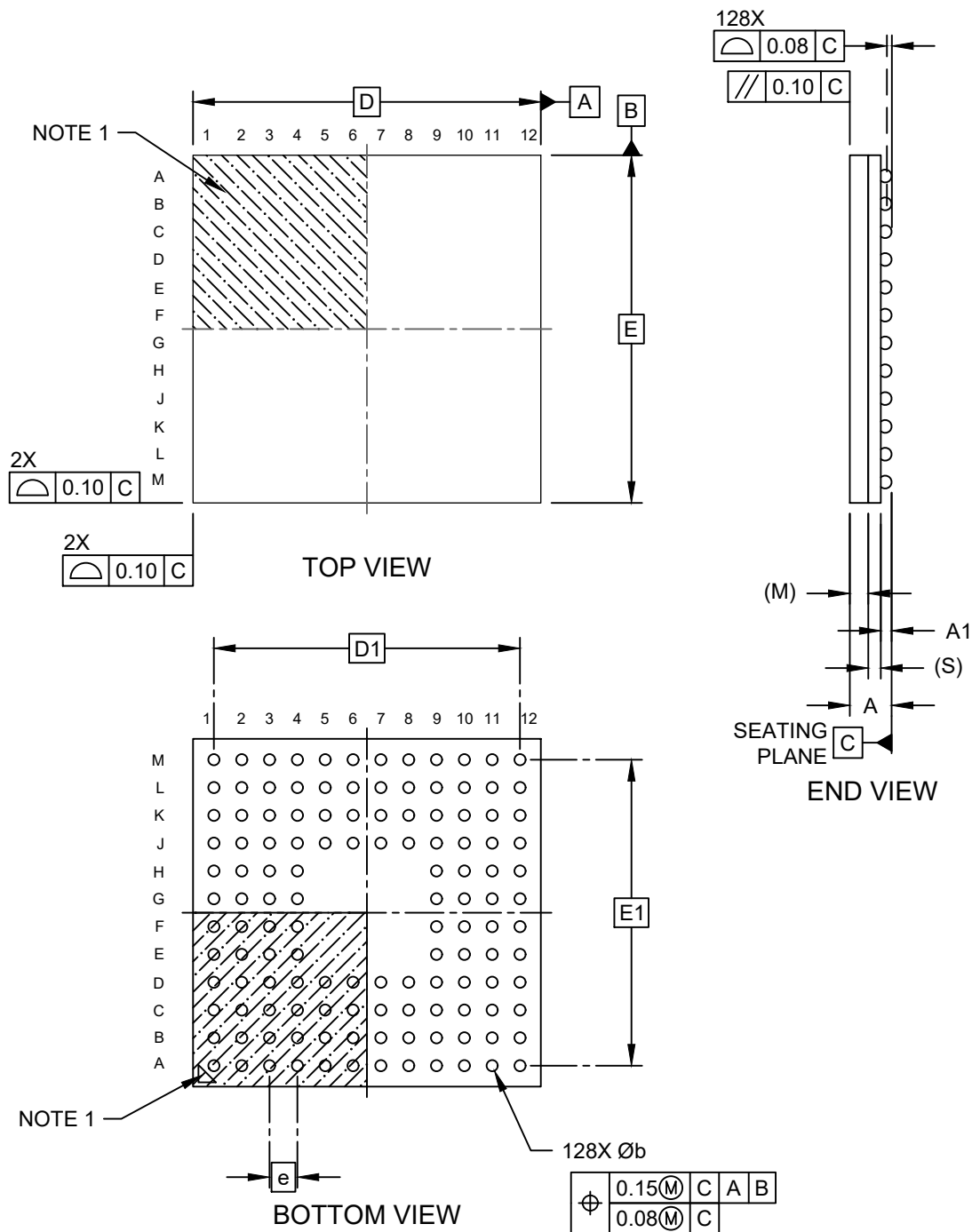
Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-23138 Rev A

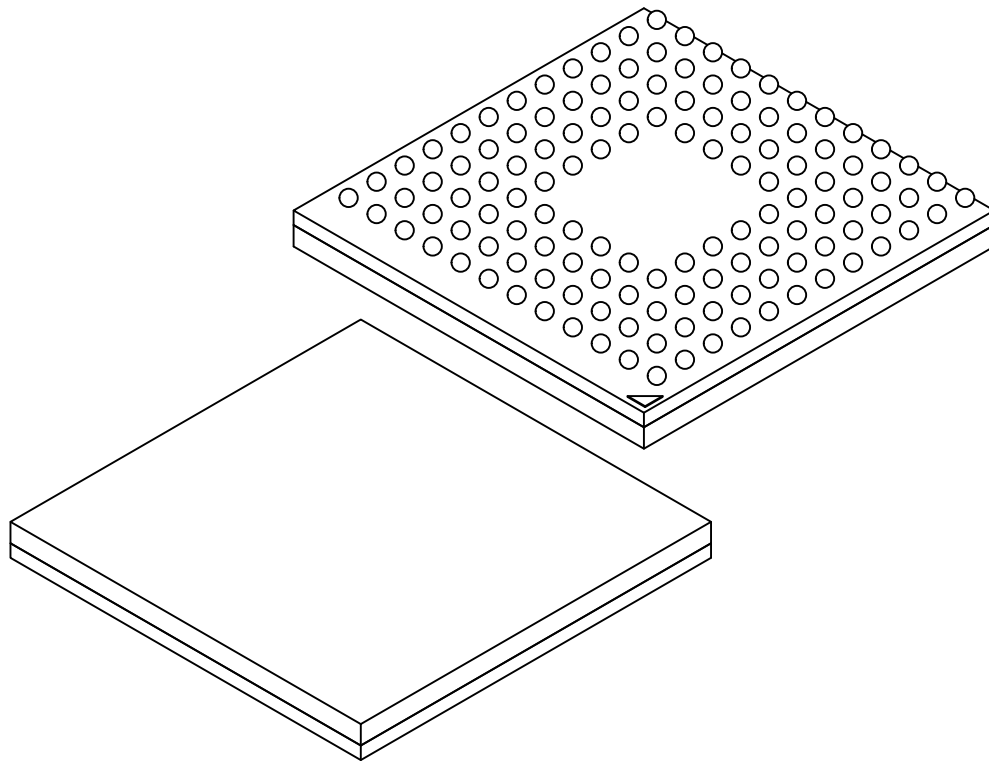
128-Ball Thin Fine-Pitch Ball Grid Array Package (B6B) - 10x10x1.2mm Body [TFBGA] Atmel Legacy GPC: CJZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



128-Ball Thin Fine-Pitch Ball Grid Array Package (B6B) - 10x10x1.2mm Body [TFBGA] Atmel Legacy GPC: CJZ

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



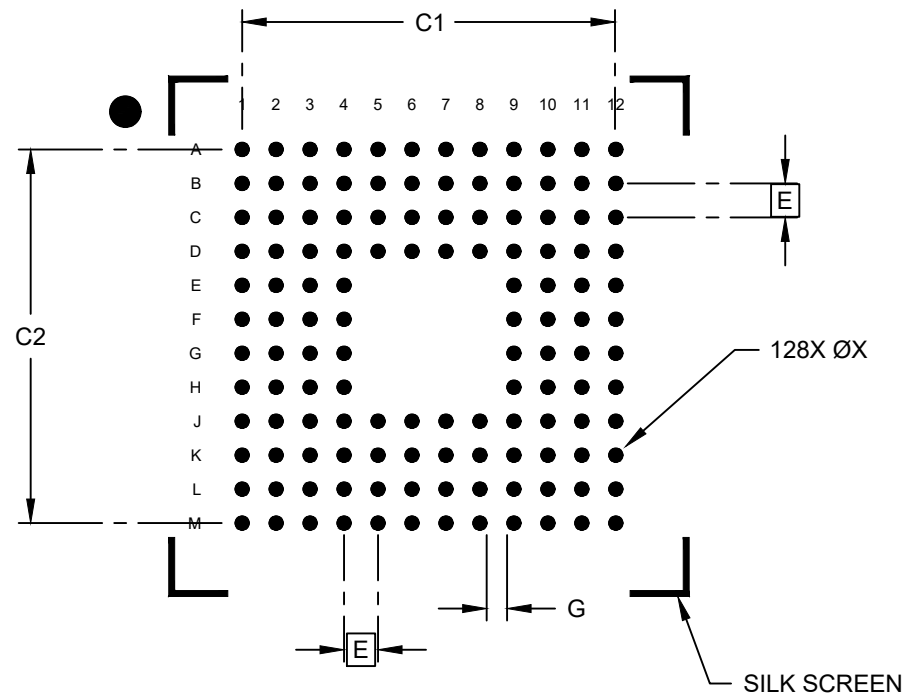
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	128		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.16	–	0.26
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.36 REF		
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Overall Width	E	10.00 BSC		
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RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Optional Pad Diameter (X128)	X			0.32
Contact Pad Spacing	C1		8.80	
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Microchip Technology Drawing C04-23138 Rev A